

EXPERIMENT – 3

DESIGNING A VIRTUAL LAN (VLAN) FOR LAB 1.a AND LAB 2.b

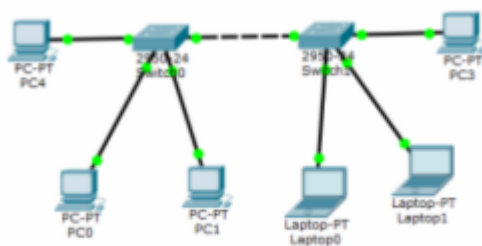
Aim

To design and configure a Virtual Local Area Network (VLAN) for AIML and CSD departments in Cisco Packet Tracer using switches, so that PCs of the same department are grouped in the same VLAN while remaining in separate broadcast domains, and to verify connectivity using PDU/ ping.

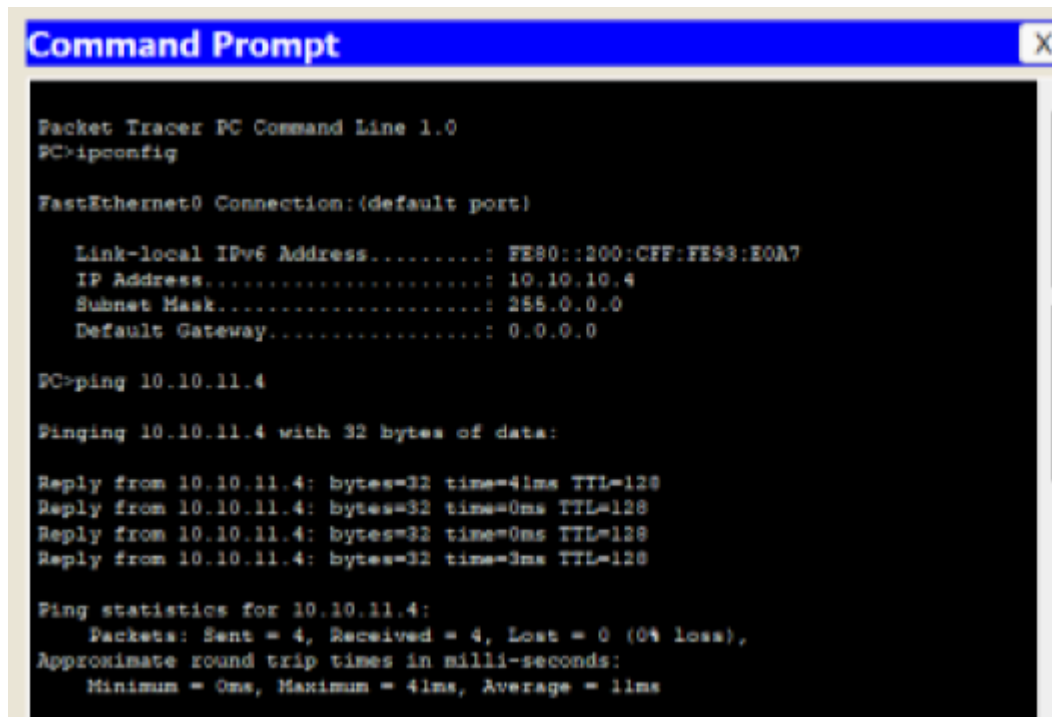
Procedure

1. Open Cisco Packet Tracer and clear the workspace.
2. Place two switches in the workspace and name them Block A and Block B.
3. Add PCs for the AIML and CSD labs (for example, two PCs for each department on each switch).
4. Connect each PC to its switch using straight through cables.
5. Connect the two switches with a cable and configure this link as a trunk port.
6. On Switch A, create VLAN 10 for AIML and VLAN 20 for CSD.
7. On Switch B, again create VLAN 10 (AIML) and VLAN 20 (CSD).
8. Assign the AIML PCs on both switches to VLAN 10 and the CSD PCs to VLAN 20.
9. Give IP addresses to all PCs so that PCs in the same VLAN are in the same IP network.
10. Enter Simulation/Realtime mode and test connectivity:
 - AIML to AIML (Block A to Block B)
 - CSD to CSD (Block A to Block B)
 - AIML to CSD (should fail)

Output



- VLAN topology diagram

A screenshot of a Packet Tracer PC Command Line window. The title bar is blue with the text 'Command Prompt' and a close button. The background is black with white text. The text shows the following commands and output:

```
Packet Tracer PC Command Line 1.0
PC>ipconfig

FastEthernet0 Connection:(default port)

    Link-local IPv6 Address.....: FE80::200:CFF:FE93:E0A7
    IP Address.....: 10.10.10.4
    Subnet Mask.....: 255.0.0.0
    Default Gateway.....: 0.0.0.0

PC>ping 10.10.11.4

Pinging 10.10.11.4 with 32 bytes of data:

Reply from 10.10.11.4: bytes=32 time=41ms TTL=128
Reply from 10.10.11.4: bytes=32 time=0ms TTL=128
Reply from 10.10.11.4: bytes=32 time=0ms TTL=128
Reply from 10.10.11.4: bytes=32 time=3ms TTL=128

Ping statistics for 10.10.11.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 41ms, Average = 11ms
```

- Ping / command prompt output

Observation

1. VLAN 10 (AIML) and VLAN 20 (CSD) were created successfully on both switches.
2. PCs in the same VLAN communicated successfully between Block A and Block B through the trunk link.
3. PCs in different VLANs were unable to communicate with each other.
4. This confirms that VLANs form separate broadcast domains in the network.

Result

A VLAN based LAN was successfully created using two switches and multiple PCs. All systems were assigned Class A IP addresses with subnet mask 255.0.0.0 in their respective VLANs, and communication between PCs in the same VLAN was verified successfully using the ping command.